Distorter: A Controllable, Real-Time Harsh-Voice Effect for Speech and Singing

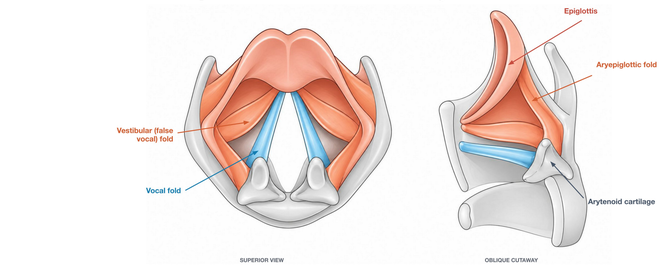
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ISMIR ABU DHABI 2026

Motivation & Background

Harsh-vocal techniques can be demanding to sustain and difficult to reproduce consistently across takes.



Physiological Background

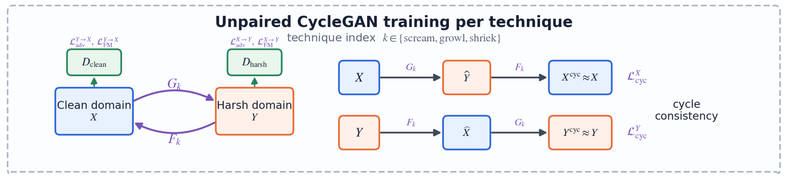
Vocal-fold phonation can be accompanied by supraglottic sound sources, producing technique-dependent spectral changes.
Aaen et al., Journal of Voice, 2024

Modeling prior: net source effects accompanying harsh phonation under shared vocal-tract shaping, approximated by a residual anchored to the input's coarse spectral envelope.
Physiology-inspired, without anatomical source estimation.

Training & Data

Unpaired training per mode

Independent CycleGANs trained from scratch without utterance-level alignment.



Training objective

$$\mathcal{L} = \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}} + \lambda_{\text{FM}}\mathcal{L}_{\text{FM}}$$

Both directions: least-squares adversarial (adv), cycle consistency (cyc), and discriminator feature matching (FM; channel-wise mean/std).

Training data

Clean domain
EMO Clean + EMVD ClearVoice for Scream.
EMVD ClearVoice for Growl and Shriek.

Harsh targets
Scream: EMO + EMVD HardcoreScream
Growl: EMVD DeathGrowl
Shriek: EMVD BlackShriek

Evaluation data

Quantitative output evaluation
VCTK speech; GTSinger / VocalSet singing.
616 inputs from 109 voices
not used for checkpoint selection.

Analysis of Real recordings
EMVD: mode features, ECAPA, SwiftF0.
EMO: Whisper on real screams.

Checkpoint validation: EMVD Singer11/12/23; Scream also uses EMO validation.

Demo examples also include outputs generated from EMVD clean recordings by singers excluded from training and checkpoint selection.

EMO: clean and vocal-fry scream recordings.
EMVD: Extreme Metal Vocals Dataset, with clean and distorted metal vocals.

Method

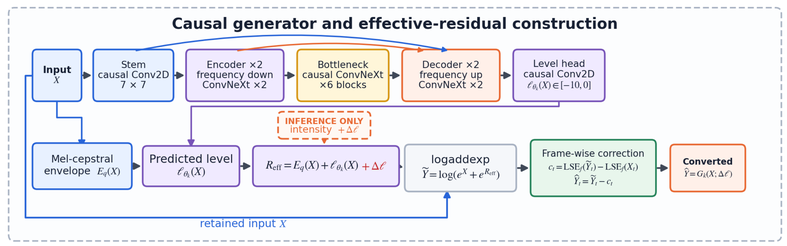
Mel-to-mel conversion

100-band log-mel magnitude spectrograms at 24 kHz.
Clean input X → converted harsh output Ŷ (log-mel).

Scream · Growl · Shriek

Three separately trained models, one per technique.

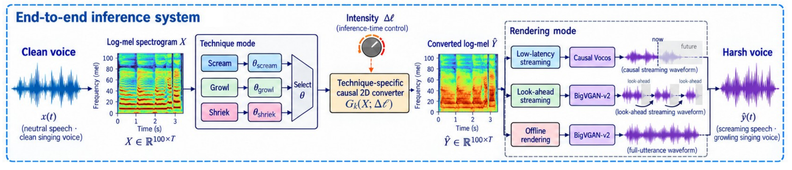
Past-only 2D ConvNeXt U-Net generators



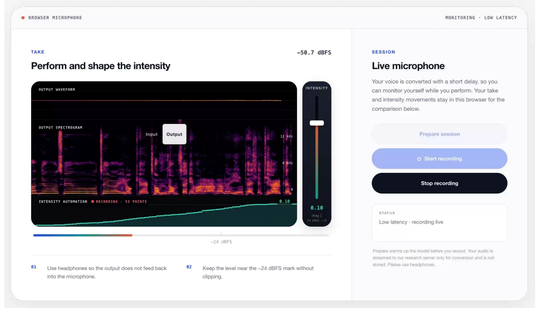
Input-envelope-anchored residual

Retain X; anchor the residual to its coarse spectral envelope.
Δℓ controls residual dominance; frame-wise correction matches input level.

System & Demo



Live Demo



1 Play a WAV
Shape intensity during playback.

2 Re-render offline
Replay the recorded intensity trajectory.

3 Live microphone
Monitor the converted voice and adjust intensity as you perform.

Results

Automatic evaluation only; no listening test.

Measurement	Result
Distinguishing the technique-targeted modes	
Separability: VCTK / GTSinger / VocalSet	78.4%, 88.1%, 75.8%
real recordings, same features	41.7% (n.s.)
Intensity control	
Suppression-depth response, −3 to +1.5 nats	$\rho = +1$, $r \geq +0.95$
peak-to-valley span, speech / singing	5–8 / 10–14 dB
frame-level matching, all / active	≤1.44 / ≤0.05 dB

14-feature classifier; balanced accuracy, chance 33.3%.
Generated: leave-one-corpus-out. Real EMVD: leave-one-singer-out.
Active: 99.99% of frames within 70 dB of the take peak.

Mode separability does not establish resemblance to real techniques.

Separability

Generated: 75.8–88.1%
Real EMVD: 41.7% (n.s.)

VCTK / GTSinger / VocalSet:
leave-one-corpus-out.
EMVD: 36 items, leave-one-singer-out.
Balanced accuracy
(chance: 33.3%).

Intensity control

Spearman ρ = +1
Pearson r ≥ 0.95

Suppression depth increases at every tested step across all 1,848 curves (Δℓ: −3 to +1.5 nats).
616 inputs × 3 modes; worst itemwise correlations, not perceived harshness.

Real-time factor < 1

Worst RTF: low-latency ≤0.63, look-ahead ≤0.35, offline ≤0.05

Configured algorithmic latency: 42.67 / 442.67 ms
(low-latency / look-ahead); not physical end-to-end latency.
Single idle NVIDIA TITAN RTX, isolated session. Excludes browser, network, scheduling/load and device buffering.

Exploratory Metric Diagnostics

Exploratory diagnostics on real harsh GT, not Distorter-output scores.

ECAPA-TDNN EER (EMVD, 9 paired lyrics): 11.1% clean → 31.5% harsh

SwiftF0 voiced coverage (EMVD, 27 paired sustained vowels):
98.6% clean → 4.6% harsh at confidence 0.90.
At confidence 0.30: 70.2% harsh, but 44.0% of jointly voiced WORLD–SwiftF0 frames differ by >300 cents.

Whisper-medium WER (EMO, 20 real screams): 69.2%

These diagnostics motivate further exploration of evaluation methods more suitable for harsh vocals.

Conclusion

Three technique-targeted modes with continuous control and offline re-rendering of intensity trajectories.

Limits: strongly harmonic inputs were least convincing in informal author listening; vocoder reconstruction errors remain.

Next: listening tests, matched baselines / ablations, and physical end-to-end latency measurements.

Listen & try

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